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Paleogene magnetic stratigraphy in Umbrian pelagic carbonate rocks: The Contessa sections, Gubbio

WILLIAM LOWRIE *Institut für Geophysik, ETH-Hönggerberg, CH-8093 Zürich, Switzerland*
WALTER ALVAREZ *Department of Geology and Geophysics, University of California, Berkeley, California 94720*
GIOVANNI NAPOLEONE *Istituto di Geologia e Paleontologia, University of Florence, Florence, Italy*
KATHARINA PERCH-NIELSEN *Geologisches Institut, ETH-Zentrum, CH-8092 Zürich, Switzerland*
ISABELLA PREMOLI SILVA *Istituto di Paleontologia, University of Milan, Milan, Italy*
MONIQUE TOUMARKINE *Geologisches Institut, ETH-Zentrum, CH-8092 Zürich, Switzerland*

ABSTRACT

The Umbrian sequence of pelagic carbonate rocks provides an opportunity for precise correlation between Paleogene biostratigraphy and geomagnetic polarity history. The red-to-pink, Paleocene to middle Eocene Scaglia Rossa limestone, the varicolored upper Eocene Scaglia Variegata limestone, and the gray-green Oligocene Scaglia Cinerea marlstone form a 250-m-thick, continuous exposure in the Contessa Valley near Gubbio, Italy. Magnetostratigraphic, lithostratigraphic, and biostratigraphic investigations in three sections covering the entire Paleogene have confirmed and dated the geomagnetic reversal sequence for most of this period. The ferromagnetic mineral in the Scaglia Cinerea is magnetite and AF demagnetization to peak fields of 20 mT is sufficient to define the characteristic remanent magnetization. The Scaglia Variegata and pink Scaglia Rossa samples contain an additional hematite component which is very pronounced in dark red Paleocene Scaglia Rossa samples. Changes of magnetic mineralogy take place in these limestones during heating, especially above 500 °C. However, thermal demagnetization is effective in isolating the characteristic remanence vectors, which form almost antipodal clusters of directions representing normal and reversed polarities. The directions are rotated counterclockwise, partly due to post-Oligocene tectonism. Paleontological zonations of planktic foraminifera and calcareous nannofossils were hampered by poor preservation, but the major epoch and stage

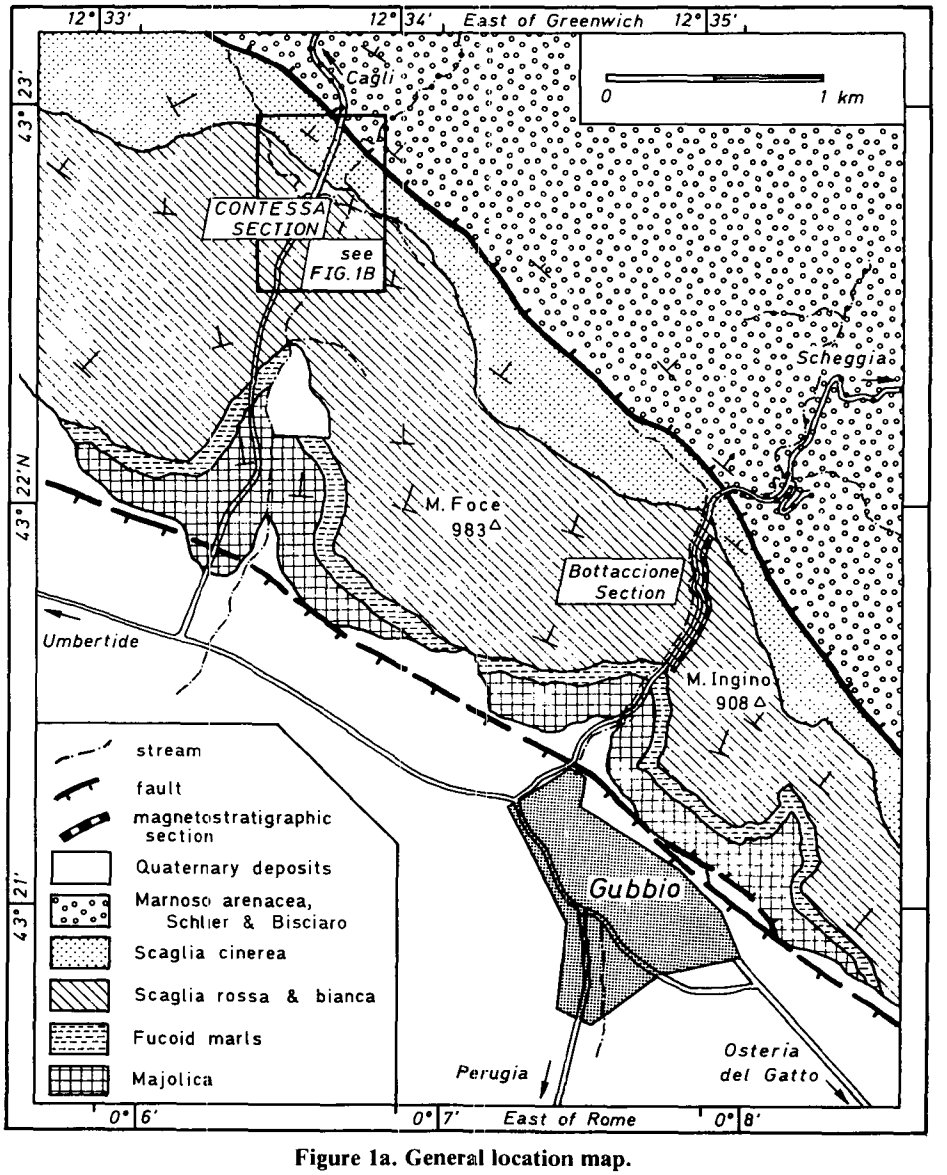


Figure 1a. General location map.

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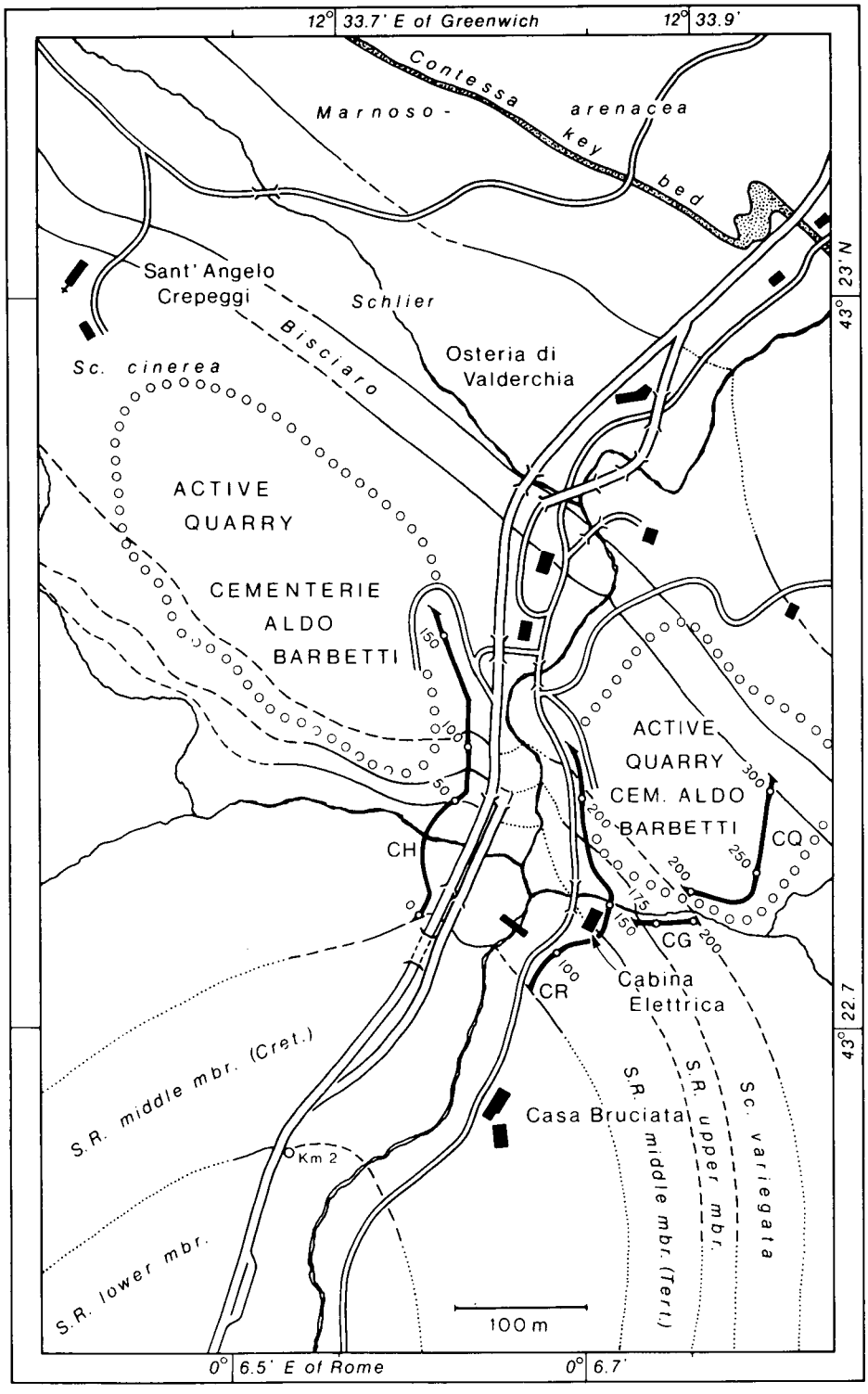


Figure 1b. Location map for the study area, showing the sampled sections: CQ and CG = Contessa Quarry, and Gully extension, CR = Contessa Road, CH = Contessa Highway.

boundaries could be satisfactorily located. Litho-, bio-, and magnetostratigraphic markers tie the three sections to each other and yield a composite magnetic reversal stratigraphy which correlates well with the Paleogene reversal sequence corresponding to oceanic magnetic anomalies 29 to 6C. The Paleogene epoch boundaries are all in negative polarity zones, as follows: Cretaceous-Paleocene, just below anomaly 29, as in the Gubbio Bottaccione section; Paleocene-Eocene, just above anomaly 25; Eocene-Oligocene, between anomalies 13 and 15; Oligocene-Miocene, just below anomaly 6C. These correlations require slight modifications to previous conclusions on Paleogene sea-floor spreading rates.

INTRODUCTION

Pelagic carbonate rocks of South Tethyan facies have suitable magnetic properties for studies of paleomagnetism and magnetic stratigraphy. Lower Cretaceous sections in the Southern Alps and in the northern Umbrian Apennines contain clearly defined magnetic reversal sequences which are easily correlated with the youngest of the M-sequence oceanic magnetic anomalies and the start of the Cretaceous magnetic quiet interval (Channell and others, 1979a; Lowrie and others, 1980a). This normal polarity zone was found in the Cismon section in the Southern Alps (Channell and others, 1979a) and in the Moria and Gubbio sections in Umbria (Lowrie and Alvarez, 1977a, 1977b; Alvarez and Lowrie, 1978); it begins at anomaly M0 at the base of the Aptian and ends at anomaly 34 near the base of the Campanian. The Upper Cretaceous and Paleocene Scaglia Rossa limestone of the Gubbio section was found to preserve a complete record of anomalies 26 to 34, with the Cretaceous-Tertiary boundary between anomalies 29 and 30 (Premoli Silva and others, 1974; Lowrie and Alvarez, 1977a, 1977b; Roggenthen and Napoleone, 1977). Because of its biostratigraphic and magnetostratigraphic completeness in the interval studied, the Gubbio section was proposed as a magnetostratigraphic type section (Alvarez and others, 1977).

In subsequent work in this section, which lies alongside the road through the Bottaccione gorge near Gubbio (Fig. 1a), Napoleone and others (in press) have extended coverage upward from upper Paleocene to upper Eocene, and are able to correlate a sharply outlined magnetic stratigraphy with Cenozoic oceanic magnetic anomalies 26 to 19. The present study was undertaken to confirm the Bottaccione results for the early Tertiary and to establish a magnetic stratigraphy in the overlying middle and upper Eocene Scaglia Variegata and Oligocene Scaglia Cinerea formations. Suitable sections were found in the Contessa gorge, about 2 km northwest of the Gubbio section in the Bottaccione gorge (Fig. 1).

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LITHOSTRATIGRAPHY AND SAMPLING

Geological Setting

The Umbrian stratigraphic sequence was deposited on the continental margin of Adria, a promontory of the African plate, on the southern side of the Mesozoic Tethys (Channell and others, 1979a). Subsidence of this crust began in the Triassic during continental rifting and continued from the onset of sea-floor spreading in the earliest Jurassic. The Umbrian sequence, mainly limestones and marls, has been described by Bortolotti and others (1970). The Jurassic through Eocene part of the sequence records the interplay between subsidence, probably due to crustal thinning (D'Argenio and Alvarez, 1980), and paleo-oceanographic events, notably changes in the calcite compensation depth (Garrison and Fischer, 1969; Bosellini and Winterer, 1975) and anoxic episodes of the mid-Cretaceous (Arthur, 1979; Arthur and Schlanger, 1979; Arthur and Premoli Silva, 1980).

The Upper Cretaceous to Eocene Scaglia Rossa is made up principally of hard, dense, pink limestone composed of about 10% foraminifera and 5% clay in a matrix of coccoliths. It has been described in detail by Arthur (1979). Three distinctive marly intervals (Marl I to III) occur in the Paleocene and the lower Eocene, and the Paleocene is marked by a reduction in sedimentation rate from about 10–15 m/m.y. in the Cretaceous to values of 1–4 m/m.y. (Arthur, 1979, Table 4). Beginning in the middle Eocene, clay input into the Umbrian area became increasingly important. The Scaglia Rossa grades upward into the gray marlstone of the Scaglia Cinerea; the transitional interval, with its alternation of pink, gray, and white beds, is sometimes designated the Scaglia Variegata and mainly corresponds to the upper Eocene. The Scaglia Cinerea marls were deposited throughout the Oligocene. Scaglia is a scaly (that is, conchoidally fracturing) limestone, but its misapplication to this marl unit is sanctioned by tradition; Cinerea refers to the ashy gray color. The Umbrian sequence closes with Miocene flysch related to the deformation of the Apennines.

The boundaries between these lithostratigraphic units have not been formally designated in the literature. For the purposes of this paper, we place the top of the Scaglia Rossa at the highest chert bed of the upper cherty member (Lowrie and Alvarez, 1975). The top of the Scaglia Variegata is placed arbitrarily in the gradational sequence at a level where gray beds become dominant. The coccolith content and preservation do not change drastically around the lithological change from Scaglia Variegata to Scaglia Cinerea. The top of the Scaglia Cinerea is an abrupt contact, with gray marl overlain by hard, resistant limestone of the Bisciaro. Baumann (1970) found a continuous faunal sequence and no evidence for a hiatus at the contact.

The lithology of the Scaglia Cinerea is briefly mentioned in a number of local and regional geological descriptions (Barnaba, 1958; Jacobacci and others, 1974). It differs by its lower $CaCO_3$ content (63% to 80%) from the Scaglia Rossa limestone (87% to 92%) (Carloni, 1962; Cati, 1964). The clay content increases slowly upward, but the uniformity of the sedimentary sequence is indicative of more or less constant depositional conditions (Baumann, 1970).

The lower to middle Miocene (Burdigalian-Langhian) Bisciaro (Guerrera, 1977) is replaced upward by the Schlier, a gray marl much like the Scaglia Cinerea. Turbidite beds soon appear, with Schlier-type hemipelagic interbeds, and this flysch unit, the Marnoso-Arenacea, closes the Umbrian sequence.

The Contessa Sections

In the Contessa valley, which cuts the same northeast-dipping, half-anticlinal structure as the Bottaccione gorge (Fig. 1a), the Scaglia Rossa, Scaglia Variegata, and Scaglia Cinerea are exposed in continuous fresh road-cuts and quarry faces several metres high. Three sections were sampled, designated informally as the Contessa Highway (CH), Contessa Road (CR), and Contessa Quarry (CQ and CG) sections (Figs. 1b, 2). The Highway and Road sections lie on opposite sides of the valley along the main highway and a minor dirt road, respectively, and cover similar stratigraphic intervals embracing the Paleocene and Eocene Scaglia Rossa and Scaglia Variegata and reaching the base of the Scaglia Cinerea. Because of its mechanically weak lithology, the Scaglia Cinerea has in many other places served as a detachment level during deformation and, as a result, most outcrops are strongly sheared and unsuitable for detailed magnetostratigraphic studies. Unusually continuous, fresh exposures occur on both sides of the Contessa valley in cement quarries of the Cementificio Aldo Barbetti (Gubbio) and are free of deformation structures except for a few minor faults and a 12.5-m interval of weak cleavage which does not interrupt the continuity or sequence of bedding. The Contessa Quarry section was placed in the active quarry on the east side of the valley and includes the entire thickness of the Scaglia Cinerea formation.

The Contessa Highway Section (Figs. 2, 3, 4) on the west side of the valley (Fig. 1b) extends 160 m from the Cretaceous-Tertiary boundary to the end of Scaglia Cinerea outcrops at the quarry entrance road. Black-on-white bull's-eyes have been painted on the outcrop at 5-m stratigraphic intervals. The three marl units of the Paleocene and lower Eocene occur in a steep lateral valley; their stratigraphic levels are not presently available. Foraminiferal zonation suggests a hiatus at about 31 m with the basal *G. velascoensis* zone missing (Fig. 4). A hiatus associated with a minor synsedimentary slump occurs at the same stratigraphic level in the nearby Bottaccione section (Napoleone and others, in press), and a comparison of the length of reversed polarity zone 24–25 in the two sections (Fig. 2) suggests that somewhat more stratigraphy is missing in the Contessa Highway section than in the Bottaccione. Above the upper cherty member, several distinctive marker beds were located: a prominent white bed followed by a set of easily eroded beds that show up as recesses and are labeled "K" through "Q." K, L, and M are distinctive triple recesses in which three 2-cm weak beds occur together in a 30-cm interval. N, O, and P are single recessed beds 3 to 4 cm thick. Q is a 50-cm recessed interval.

The Contessa Road Section (Figs. 2, 5, 6) begins at the Cretaceous-Tertiary boundary, but because of the presence of a substantial fault which obscures lower Eocene Marl III, paleomagnetic sampling was not carried to the base. The Road section agrees very closely with the Highway section in the thickness of the upper cherty member, and in the placement of the white marker and beds K through Q, which are unequivocally recognizable. The section terminates upward at the end of the outcrop.

The Contessa Quarry Section (Figs. 2, 7, 8) was extended downward into an adjacent gully (CG, Fig. 1b) to connect and overlap with the Road and Highway sections. This west-flowing gully drops over a sheer cliff above the Cabina Elettrica in the Road section (Fig. 1b), but the top of the upper cherty member can be accurately traced up the cliff to the start of the Gully extension (CG). Markers L, M, and N were tentatively identified, and subsequently the magnetic

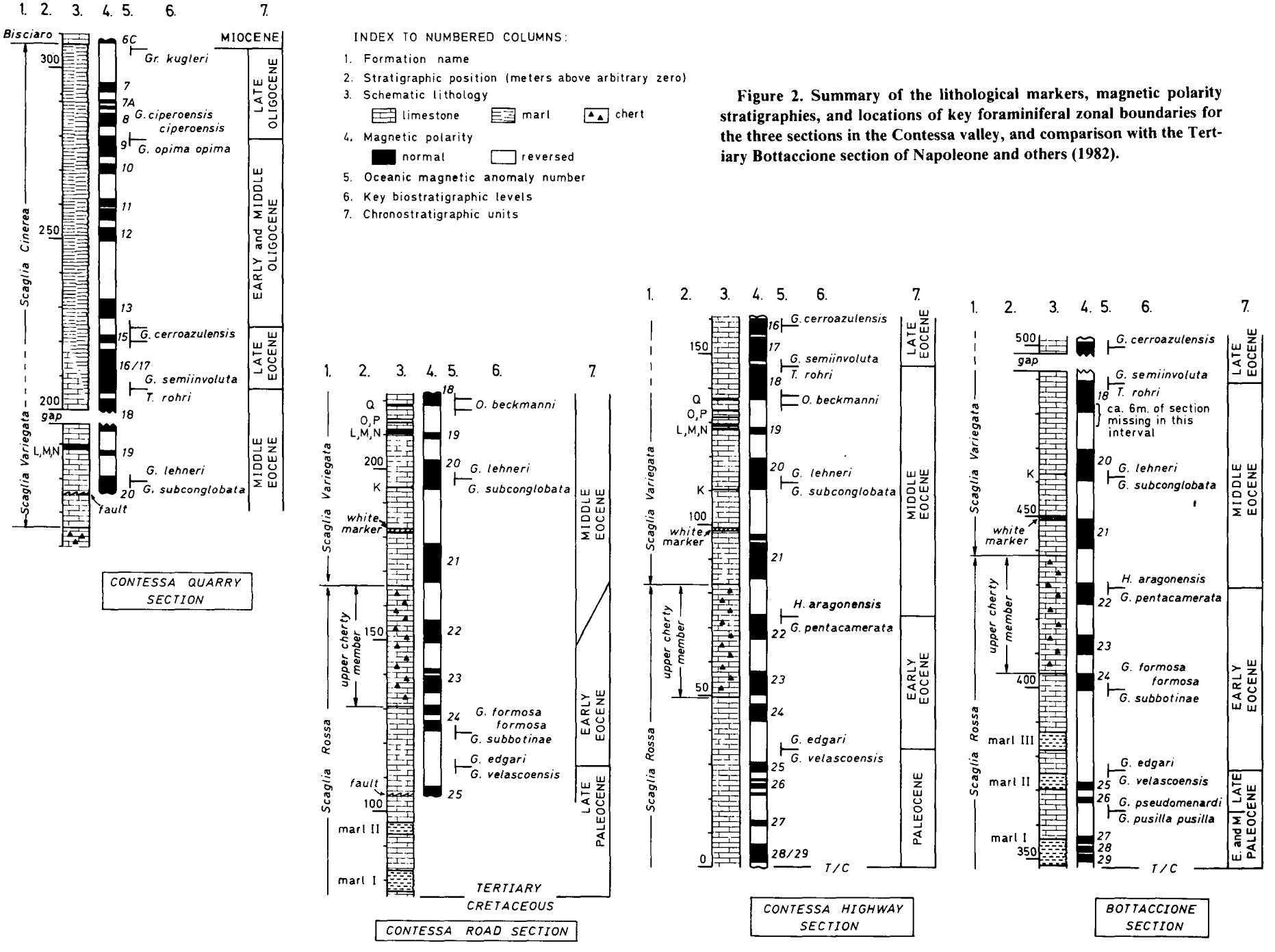


Figure 2. Summary of the lithological markers, magnetic polarity stratigraphies, and locations of key foraminiferal zonal boundaries for the three sections in the Contessa valley, and comparison with the Tertiary Bottaccione section of Napoleone and others (1982).

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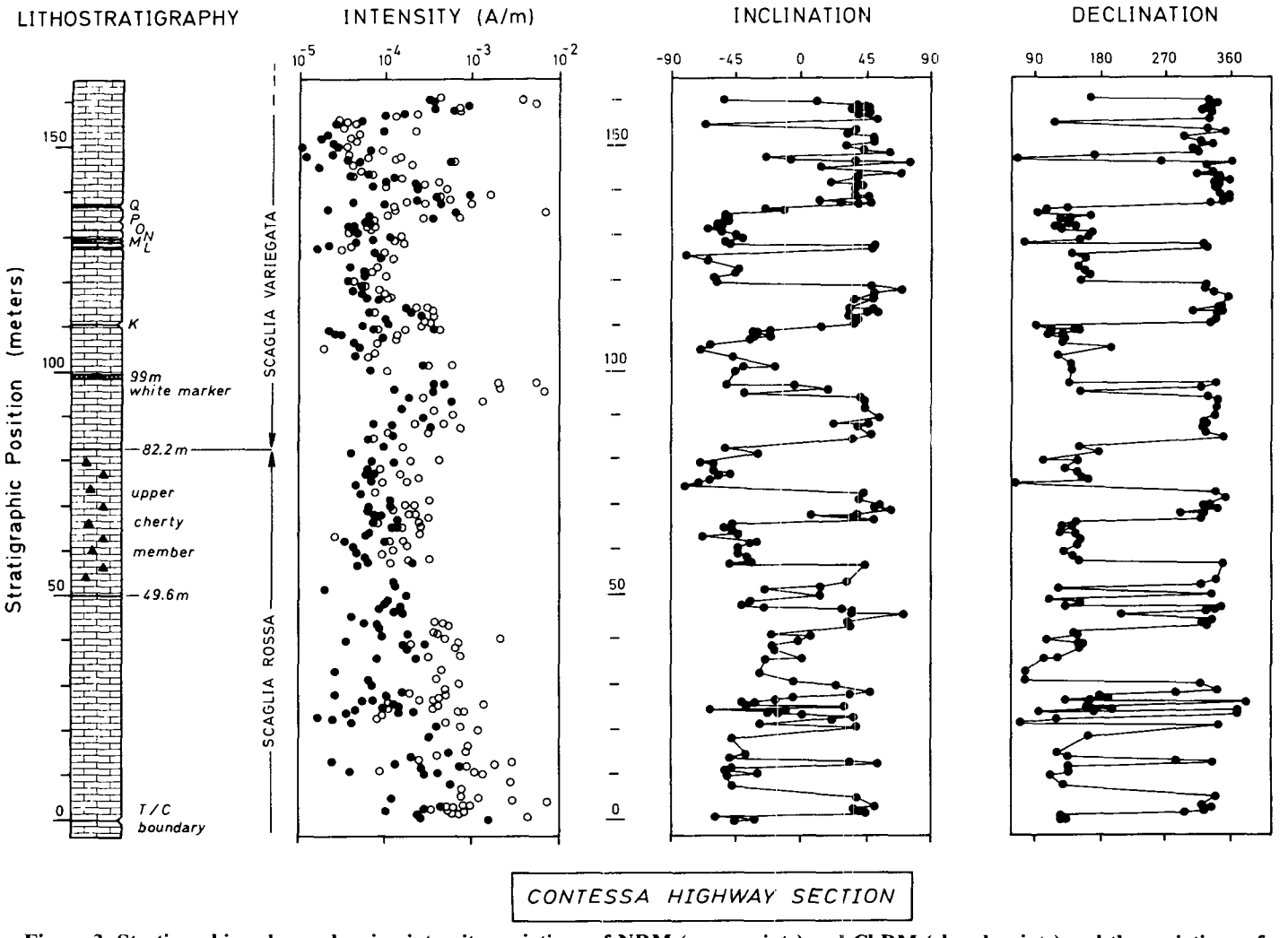


Figure 3. Stratigraphic column showing intensity variations of NRM (open points) and ChRM (closed points) and the variations of inclination and declination of ChRM in the Scaglia Rossa and Scaglia Variegata of the Highway section.

zonation showed this identification to be correct. The absence of the white marker and the reduction in stratigraphic thickness between the highest chert and marker L show that about 21 m are missing from the Gully part of the section, and, indeed, faults of uncertain throw are present in this interval.

Spill from quarrying operations covers the north slope of the gully, resulting in a 4-m gap between the Gully and Quarry sections (Figs. 7, 8). Fault planes are visible in the upper part of the gully and in the lower part of the quarry, and we suspected faulting under the covered gap between the sections, but the magnetic zonation (Figs. 2, 8) ties closely to the marine magnetic anomaly sequence (Fig. 9), suggesting that the gap is free of faults. In the Quarry section, we were unable to find a single distinctive marker bed in the monotonous gray marls of the Scaglia Cinerea.

Renz (1936; 1951, p. 27, 37, 38) reports that Bonarelli found a level in the upper part of the Scaglia Cinerea in the Bottaccione gorge containing macrofossils—oysters and *Acesta,* a pecten-like pelecypod. Bonarelli (1901) notes the occurrence of these forms in the Cinerea in unspecified outcrops. During the present study, W. A. and Ettore Sannipoli of Gubbio found several specimens of *Acesta* (identified by Carole S. Hickman) in the interval 292–294 m of the Quarry section. Unfortunately, these fossils do not have depth significance; living forms of *Acesta* range from sea level to 2,660 m depth (Kauffman, 1964).

Five stratigraphic levels between 214 and 232 m in the Quarry section are marked by scatters of biotite grains. The fact that these grains are confined to distinct levels suggests that they were emplaced as air-fall volcanic ash. To our knowledge, this is the first indication of a volcanic component in the Scaglia Cinerea from this area.

The locations of numerous small faults in the section were noted (Fig. 7). Several occur in the interval 235.5–256.3, but these faults apparently have very minor throw. A small fault at 222.35 is at the top of a 20-m interval where the sequence may be slightly disturbed by bulldozing, and where the polarity zone–anomaly match is not perfect. From 293.5 to 306.0 m, there is a zone in which the marls show incipient spaced cleavage at a high angle to bedding. There is a preferred alignment of fine phyllosilicates on the cleavage surfaces, which probably represent a very weak solution cleavage (Alvarez and others, 1976; Alvarez and

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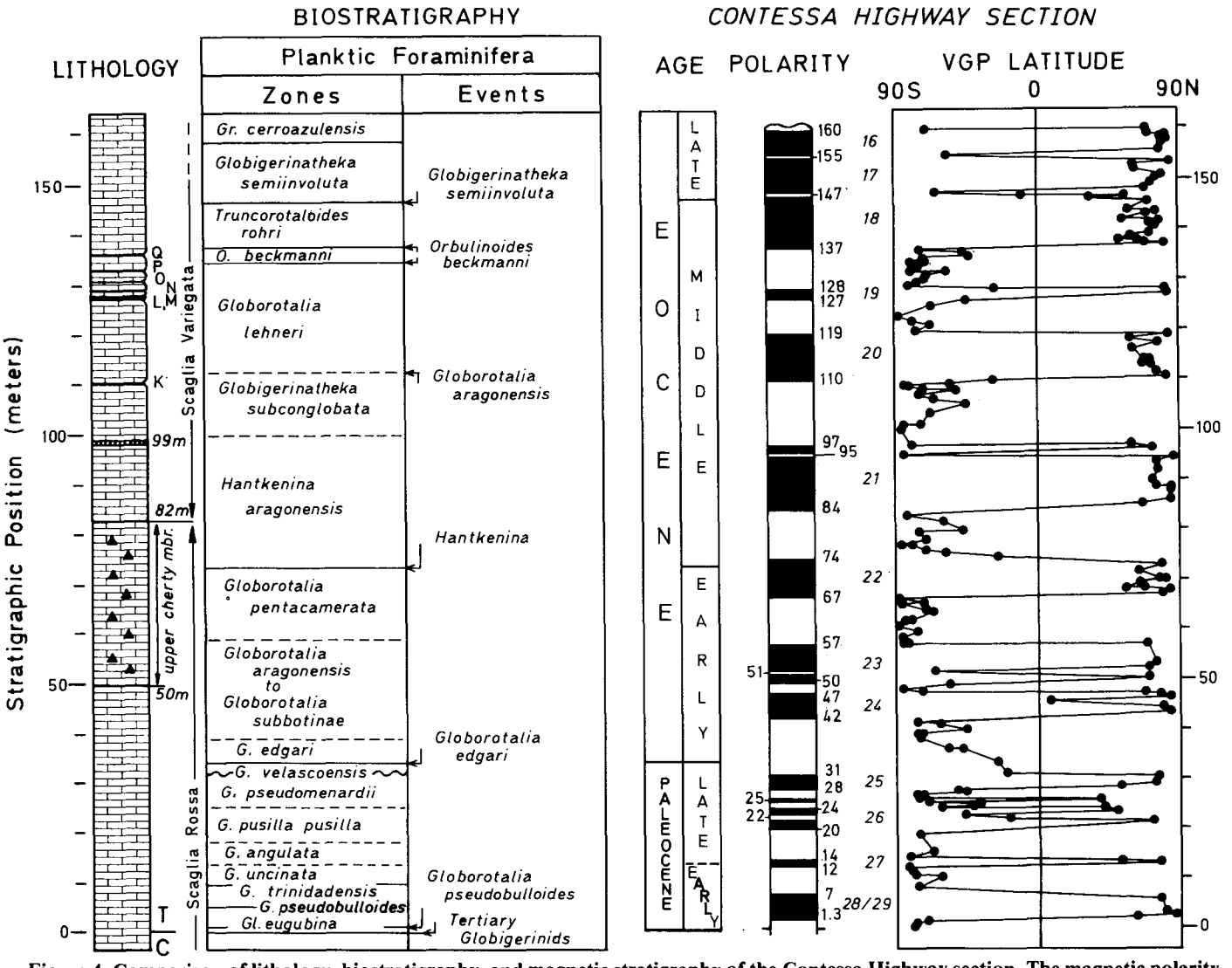


Figure 4. Comparison of lithology, biostratigraphy, and magnetic stratigraphy of the Contessa Highway section. The magnetic polarity column is derived from the relative VGP latitude plot (see text). Numbers on the polarity column indicate reversal boundary position to nearest metre (relative to arbitrary zero level); the corresponding oceanic magnetic anomaly number is indicated for each magnetozone. Stratigraphic position in this section is relative to the Cretaceous-Tertiary boundary.

others, 1978). The phyllosilicate alignment imparts a sheen by which this zone can be recognized at a distance. Measurement was terminated upward at the resistant limestones which mark the base of the Bisciaro formation (Guerrera, 1977).

The Bottaccione Section

Figure 2 also shows, for comparison, the Tertiary portion of the Bottaccione section, 2 km southeast of the Contessa Valley (Fig. 1a). Detailed studies of the Bottaccione Tertiary have been published by Premoli Silva (1977), Roggenthen and Napoleone (1977), and Napoleone and others (in press). The three marl units of the Paleocene and lower Eocene are present. Magnetic stratigraphy (Fig. 2) suggests that Marl II is of different ages in the Bottaccione and Contessa Road sections; this seems unlikely in view of their proximity and will merit further investigation. Higher in the Bottaccione section, above the upper cherty member, the white marker and bed K were tentatively identified and subsequently confirmed by magnetic stratigraphy. They clearly match the CH and CR sections, although the interval from the highest chert to the white marker is about 5 m thinner in the Bottaccione section. Markers L through Q could not be found in the Bottaccione section. The covered gap in the section at 492–497.5 m is too thin to hide the marker sequence, which has probably been removed by the faulting in the interval 477–483 m that was inferred by Napoleone and others (in press) to explain a gap in the magnetic and biostratigraphic zonations.

Paleomagnetic and Paleontological Sampling

At this location, the Oligocene is about 82 m thick, the Eocene is about 135 m thick, and the Paleocene is about 35 m thick. The sedimentation rates deduced from the paleontological dating and the Paleogene

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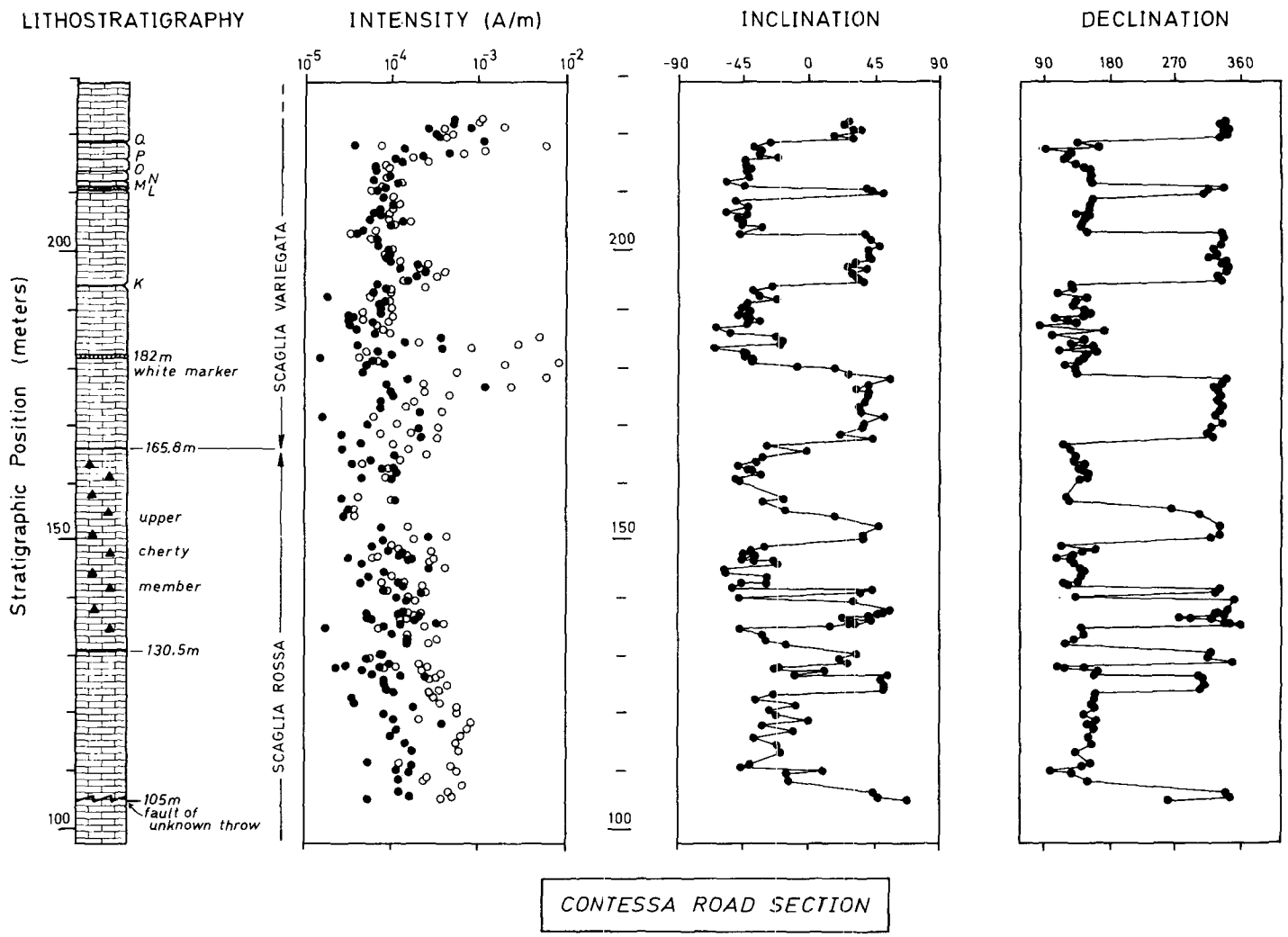


Figure 5. Lithostratigraphy, stratigraphic variations of the intensities of NRM (open points) and ChRM (closed points), and the inclinations and declinations of ChRM in the Contessa Road section. K is a marl triplet and L–Q form a noticeable pattern of marly beds; the white marker bed and the upper and lower bounds of the upper cherty member of the Scaglia Rossa are important for correlation to the other sections.

numerical time scale of Hardenbol and Berggren (1978) are about 6 m/m.y. in the Oligocene Scaglia Cinerea, 9 m/m.y. in the Eocene Scaglia Variegata and Scaglia Rossa, and 3 m/m.y. in the Paleocene Scaglia Rossa. In order that the resolution be equivalent to about 10^5 years or better, a paleomagnetic sample spacing of about 30–60 cm was required. The marly Scaglia Cinerea was very friable; in places where it could not be sampled with a portable drill (for example, in the zone of strong cleavage between 293.5 m and 306 m in the Quarry section), small oriented chips were taken and later shaped into measurement cylinders in the laboratory. Similar problems were encountered with the fissile shaly samples of the Paleocene Scaglia Rossa, which frequently had to be glued. The rest of the Scaglia Rossa and Scaglia Variegata presented no sampling difficulties. A total of 558 samples, yielding more than 700 specimens, were obtained for the rock magnetic and paleomagnetic studies (Table 1).

Paleontological samples were taken at about 2-m intervals in the friable Scaglia Cinerea. It was more difficult to obtain suitable samples in the Scaglia Variegata and Scaglia Rossa. Samples to be washed were taken at rather irregular intervals in the Contessa Road and Contessa Highway sections. The planktic foraminiferal zonation of the latter section was enhanced by observations in thin sections, at 50-cm spacing in some intervals.

The Road and Highway sections complemented each other and could be resampled conveniently where necessary for refinement of the biostratigraphy or magnetic stratigraphy. However, the measured Quarry section was destroyed by subsequent quarrying operations, and later resampling was not possible.

MAGNETIC PROPERTIES

Isothermal Remanent Magnetization (IRM) and Susceptibility

The magnetic mineralogy of the limestones was investigated by coercivity spectrum analysis (Dunlop, 1972). Thirty unheated samples were given IRM in progressively increasing magnetizing fields up

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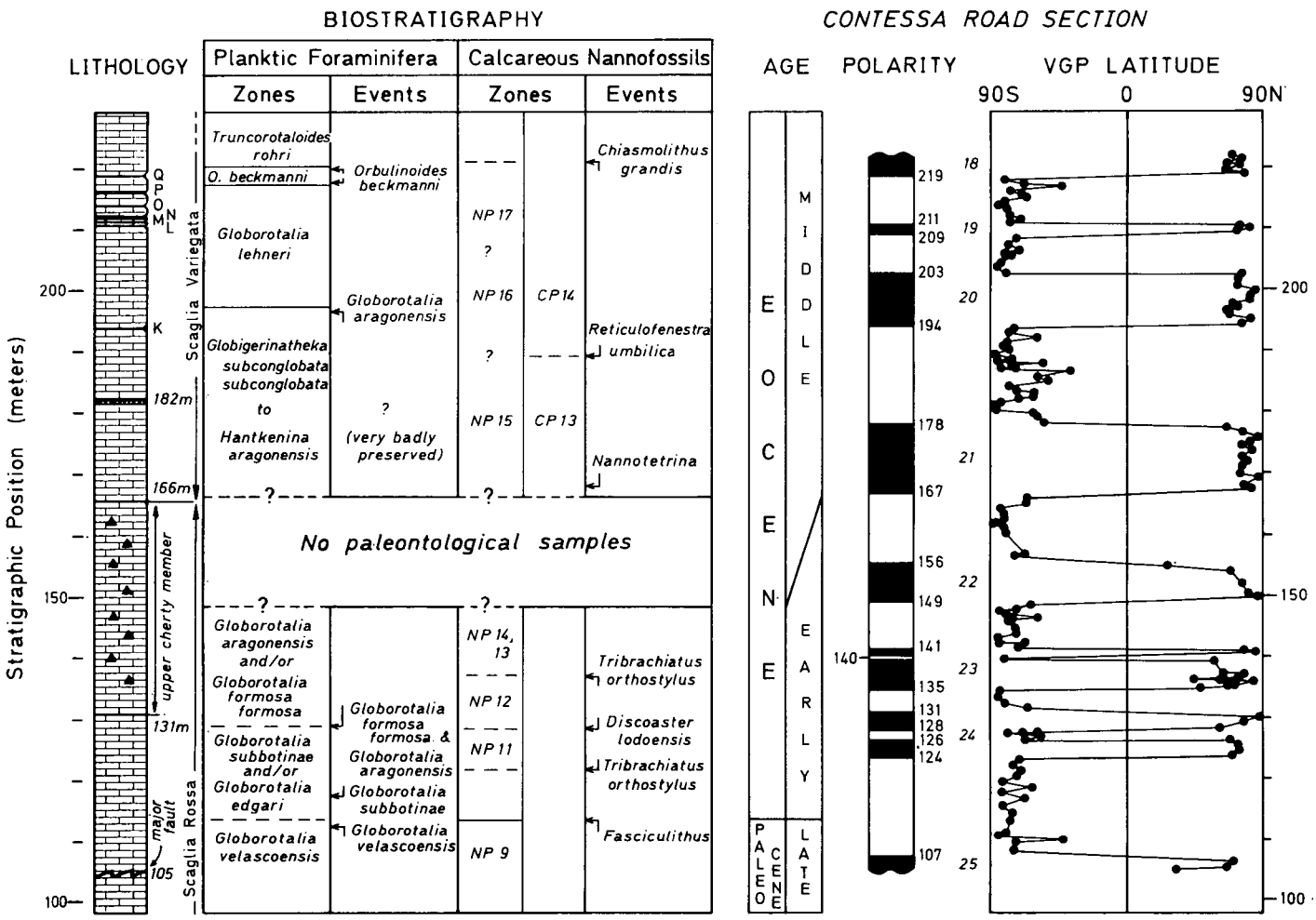


Figure 6. Lithology, biostratigraphy, and magnetic stratigraphy of the Contessa Road section. The calcareous nomenclature NP is from Martini (1971), and the CP zonation is from Okada and Bukry (1980) (for further explanation, see Fig. 4).

to 1.2 T with an electromagnet, or up to 5 T with a cryogenic magnet. Smooth IRM acquisition curves were obtained; these were used to calculate the total coercivity spectra of the samples (Fig. 10). In every sample, the largest part of the coercivity spectrum lies below 0.2 T and can probably be associated with magnetite. The Scaglia Cinerea samples all reached effective saturation IRM in about 0.2 T. However, many of the Scaglia Variegata and Scaglia Rossa samples reached saturation IRM only after applying fields of about 1.5 T. The high coercivity component in the Variegata and Rossa samples is probably associated with hematite.

Prominent visible and magnetic alterations occurred during high-temperature heating of Cinerea and Variegata samples. The gray-green Cinerea samples changed to a tan color and frequently disintegrated during heating, sometimes explosively. The changes reflect dehydration of the clays at about 400–500 °C, with associated build-up of pressure within pores or cracks as the water content is driven off. The Variegata samples also changed appearance as a result of high-temperature heating; after heating above 600 °C they took on a definite reddish hue. The Scaglia Rossa samples showed little effect of heating until 650 °C, above which temperature the red color bleached out.

Susceptibilities of all samples were monitored at room temperature after each heating stage and demonstrated that magnetic alterations accompanied the visible effects of heating at high temperature (Fig. 11). In Scaglia Cinerea samples, the susceptibilities were of an order of magnitude higher than in Scaglia Variegata samples. This difference matches the difference in NRM intensities and reflects a larger detrital component in the marly Cinerea than in the Variegata limestones. Above 350–400 °C the Cinerea susceptibilities increased, most prominently above 450 °C. A slight reduction in the Variegata susceptibilities took place above 200 °C; above temperatures of 500–550 °C a large increase in susceptibility resulted. In many Cinerea and Variegata samples, fourfold increases in susceptibility were recorded.

The IRM coercivity spectra were affected by the high-temperature changes (Fig. 11). Most significant is the dramatic increase in the fraction up to 4 kOe; no appreciable change in higher coercivity components was detectable. A new, highly magnetic phase (probably magnetite) was apparently generated by the breakdown of clay and/or sulfide minerals.

The Paleocene Scaglia Rossa samples

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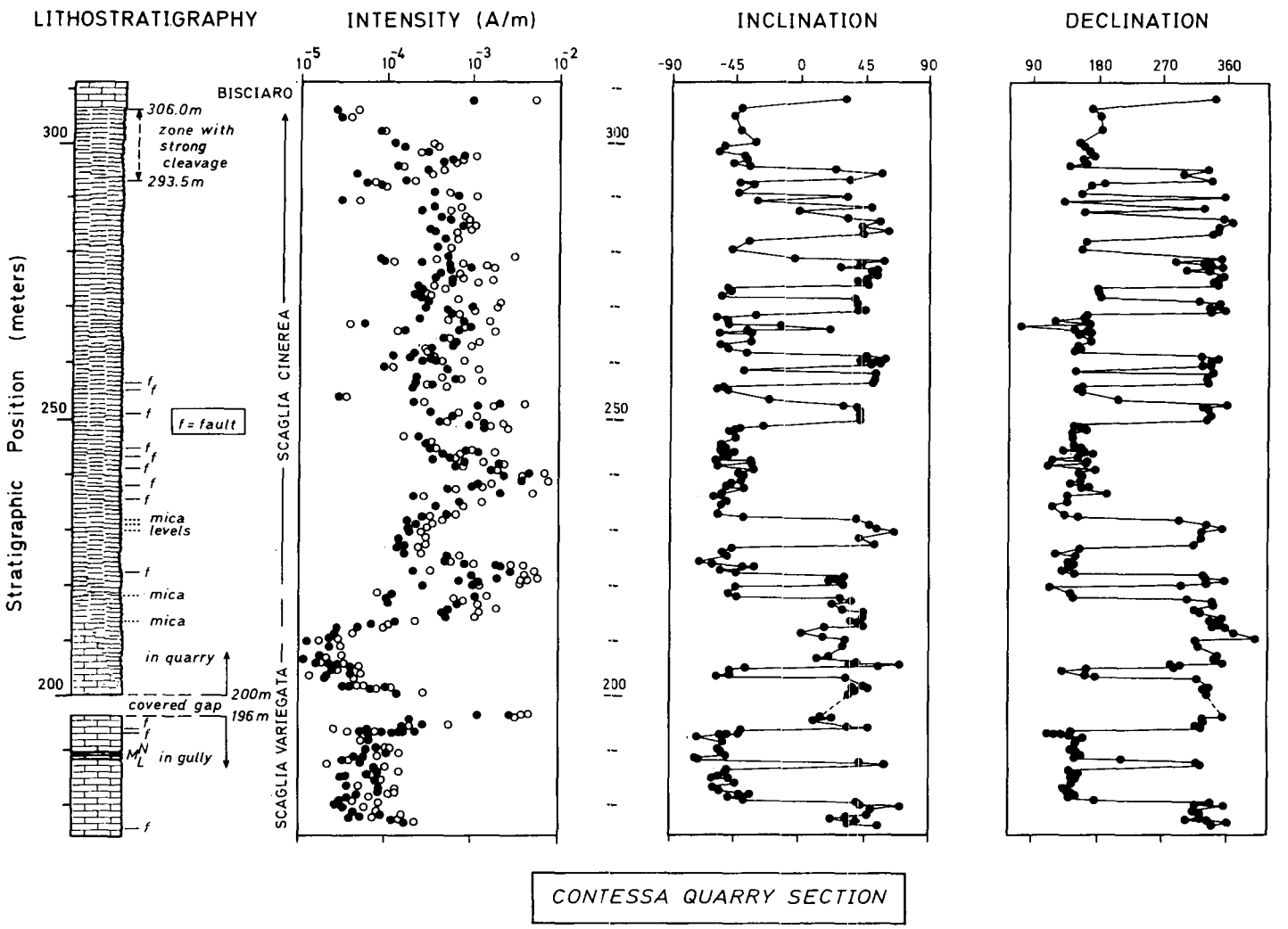


Figure 7. Magnetic stratigraphy of the Contessa Quarry section (for explanation, see Fig. 3). A 4-m covered gap at ~200 m separates samples from the quarry proper and from the adjacent gully.

were quite dark red in color. The initial susceptibilities decreased slightly after heating above 300–400 °C (due probably to destruction of an unstable mineral) and increased only after heating to 550–600 °C (Fig. 12).

The Origin of the Natural Remanent Magnetization (NRM)

Studies on other Scaglia limestones (Lowrie and Alvarez, 1975, 1977b; Channell, 1978) have assumed a postdepositional remanent magnetization mechanism in which minute magnetite grains are oriented statistically by the ambient magnetic field in water-filled interstices soon after deposition (Irving and Major, 1964). The intensity of remanence is determined by the ambient field intensity and the concentration of magnetic minerals in the sediment; the direction of remanence depends on the field direction, but is independent of the magnetic mineral concentration and, therefore, of lithological variations. The hematite in Umbrian Scaglia forms during early diagenesis, probably from a goethite precursor (Channell, 1978), and acquires its remanent magnetization almost contemporaneously with the magnetite component (Lowrie and Alvarez, 1975).

Measurement and Analysis of the Natural Remanent Magnetization (NRM)

The remanent magnetizations were measured with a 3-axis ScT cryogenic magnetometer (Goree and Fuller, 1976). The quality of each measurement was assessed from the repeatability of each magnetization in three or six measurement positions (Lowrie and others, 1980b). Remanences weaker than 1×10^{-5} A/m were usually too poorly defined to be reliable and were rejected. The NRM intensities averaged 66×10^{-5} A/m in the Scaglia Cinerea, 14×10^{-5} A/m in the Scaglia Variegata, and 39×10^{-5} A/m in the Scaglia Rossa.

The stability of natural remanence in all three lithologies was studied in seventy pilot specimens (representing 10% of the collection) selected at ~4-m intervals in the three sections. One-third were cleaned magnetically by progressive alternating field (AF) demagnetization with a two-axis tumbler.

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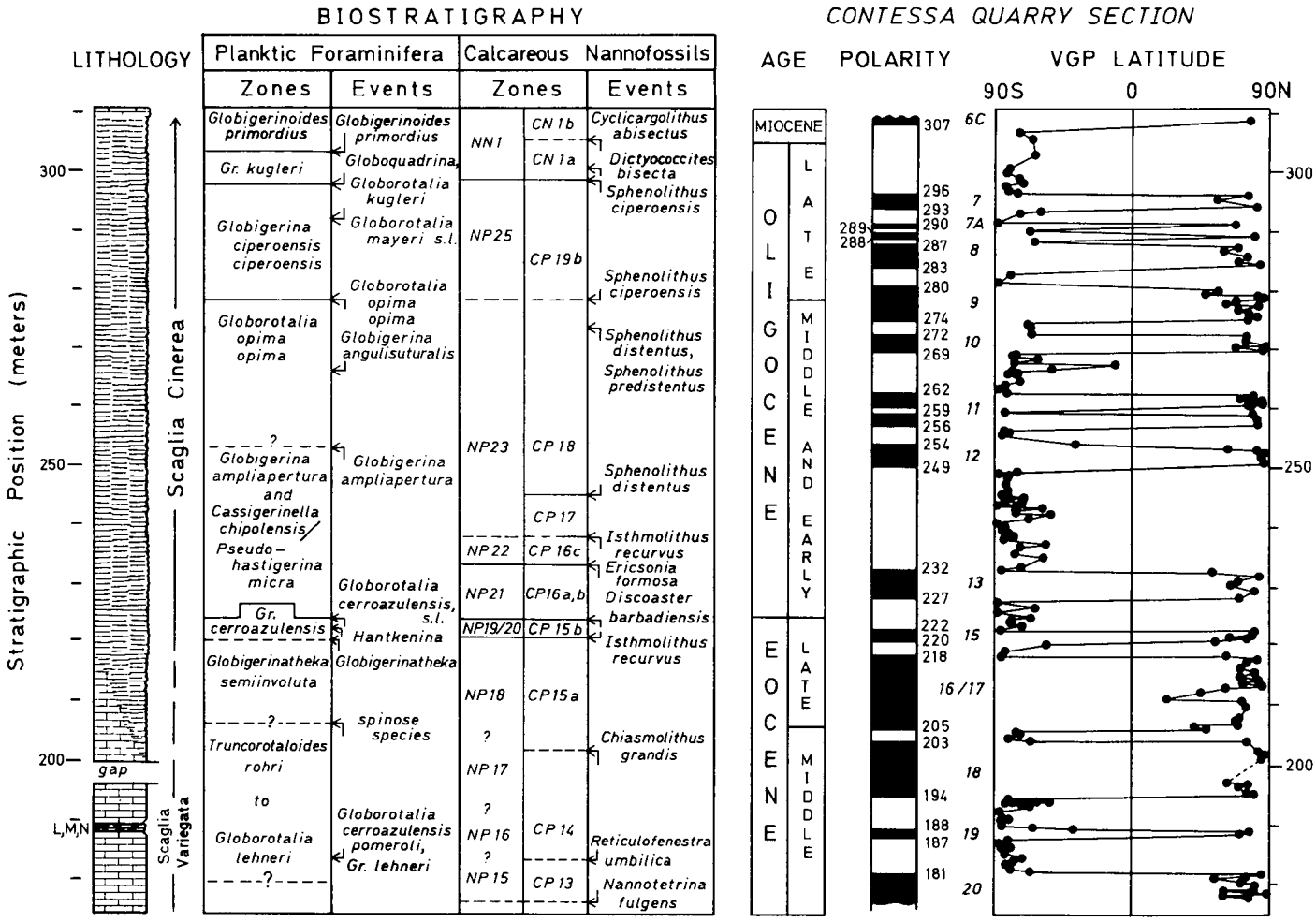


Figure 8. Lithology, biostratigraphy, and magnetic stratigraphy of the Contessa Quarry section (for further explanation, see Fig. 4).

Peak field increments of 5 mT were used up to 40 mT followed by 10 mT increments up to 100 mT; anhysteretic or rotational remanence effects were observed only above 80 mT. The remaining two-thirds of the pilot samples were thermally demagnetized in 50-degree steps up to 450 °C and in smaller steps up to a maximum temperature of 600 °C or 700 °C.

Alternating field treatment was adequate for most of the Scaglia Cinerea samples. Above 10 mT, the magnetic vector decayed smoothly and uniformly; the characteristic remanent magnetization (ChRM) direction was defined by a straight line to the origin of a vector diagram (Fig. 13). The Scaglia Variegata and Scaglia Rossa samples were less efficiently cleaned; low fields removed an unstable component and higher fields defined a vector which decayed toward the origin but did not reach it by 80 mT.

Thermal demagnetization was effective in cleaning all the Scaglia lithologies. An initial unstable component was removed by 200 °C in the Cinerea, 300 °C in the Variegata, and 400 °C in pink Scaglia Rossa samples (Fig. 14.) The ChRM was clearly defined above these temperatures. In a few Cinerea samples and most of the Scaglia Variegata and Scaglia Rossa samples, a small component persisted after heating above 600 °C. In all the Cinerea and some of the Variegata samples, thermal demagnetization was complicated by (1) creation of a new, strongly magnetic mineral during the heating, as indicated by the rock magnetic studies, and (2) deterioration of the samples during heating above 500 °C, sometimes leading to explosive disintegration.

During thermal demagnetization of red Paleocene Scaglia Rossa samples (Fig. 12), components removed below 350 ° have directions nearly parallel or antiparallel to the present field; the origin of these secondary components, acquired after the folding of the Apennines, is not clarified as yet, but they evidently developed during more than one field polarity. Above 400 °C in most of these samples, but 500 °C in some cases, the ChRM vector was defined.

Characteristic Remanent Magnetization (ChRM)

In view of the demagnetizing behavior of the pilot specimens, slightly different magnetic cleaning treatments were used for each lithology. All Scaglia Cinerea samples were cleaned by AF demagnetization in 10 mT and 20 mT, followed by thermal demagnetization at 350 °C and 450 °C. Scaglia Variegata samples were thermally demagnetized

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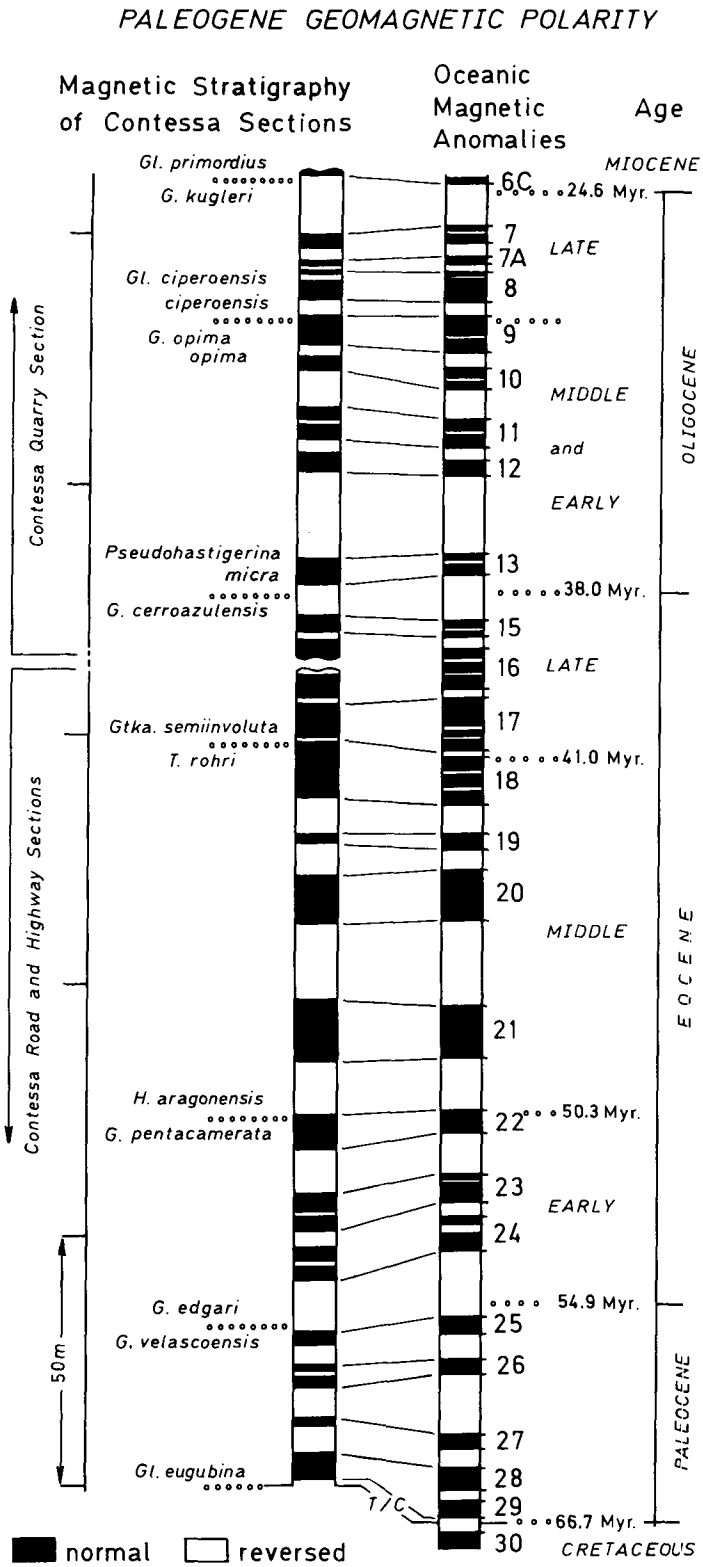


Figure 9. Comparison of the magnetic polarity stratigraphy of the Contessa Quarry section and the composite of the Contessa Road and Highway sections with the Paleogene geomagnetic polarity sequence inferred from oceanic magnetic anomalies by LaBrecque and others (1977). The numeric ages of the Paleogene stage and epoch boundaries are from Ness and others (1980).

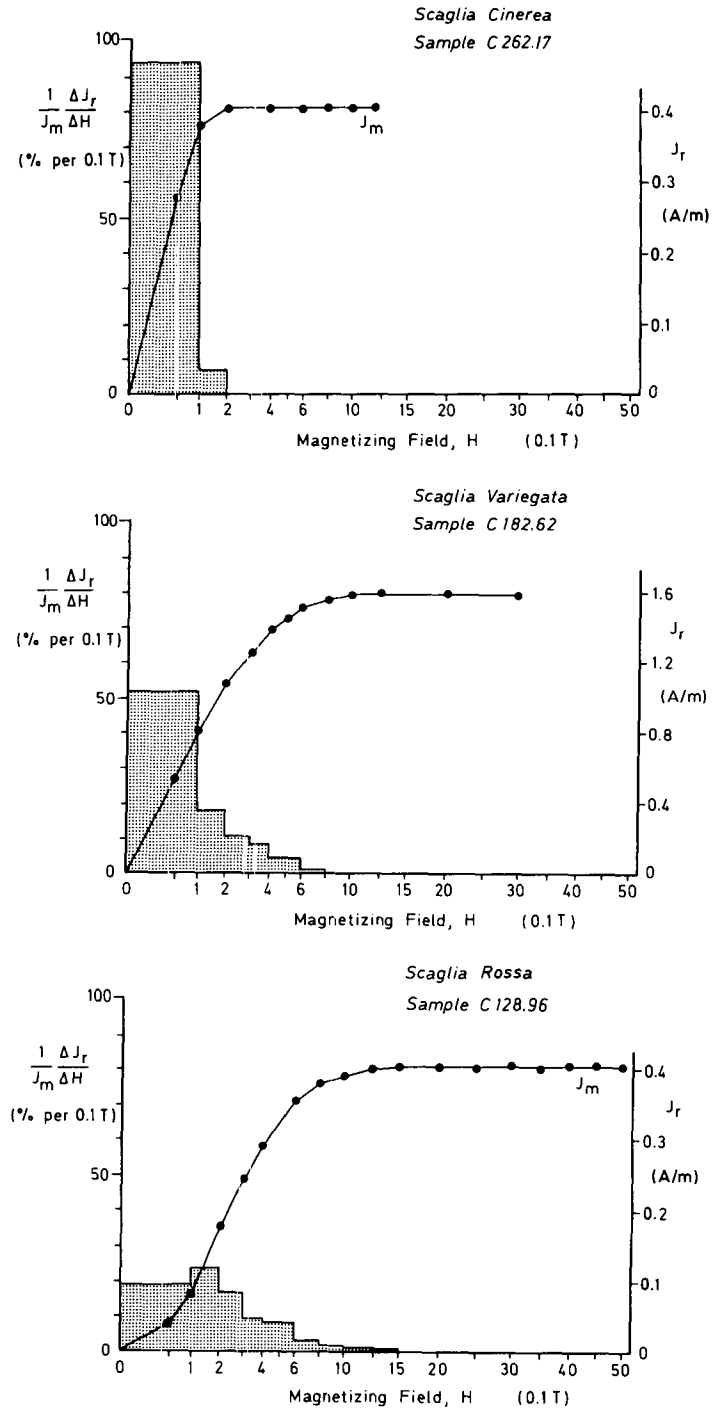


Figure 10. Acquisition of isothermal remanent magnetization (IRM) in each of the three lithologies investigated. The Scaglia Cinerea magnetic mineral is magnetite. This is also the dominant component in the Scaglia Variegata and Scaglia Rossa, in which the higher coercivity components are associated with hematite.

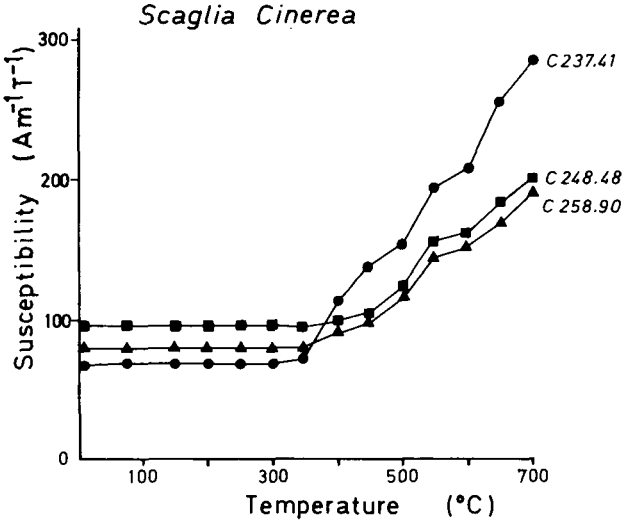
Figure 11. Room-temperature monitoring of susceptibility after step-wise heating to high temperatures. Dramatic increases of susceptibility indicate growth of a new magnetic phase above 400 °C in the Scaglia Variegata. IRM curves and coercivity spectra $1/J_m \cdot \Delta J_r/\Delta H$ (histograms) before and after heating to 700 °C indicate that the new phase has low coercivity and is probably magnetite.

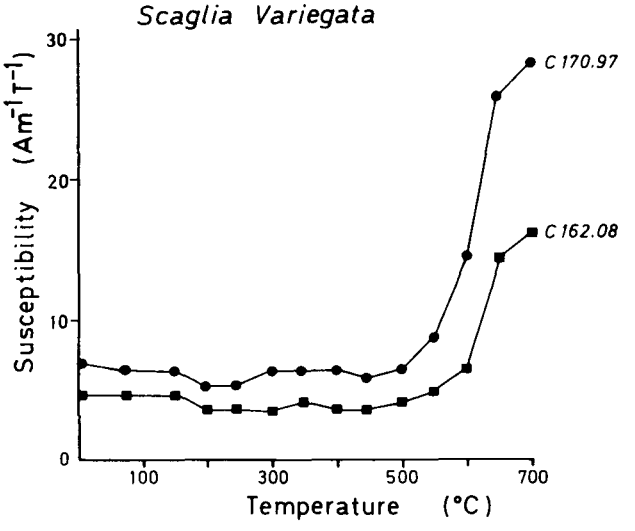
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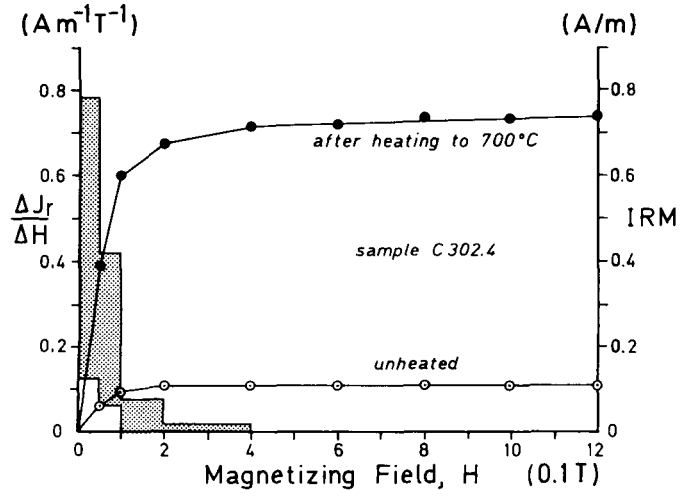
TABLE 1. SUMMARY OF CONTESSA SECTIONS
PALEOMAGNETIC DIRECTIONS AND VGP

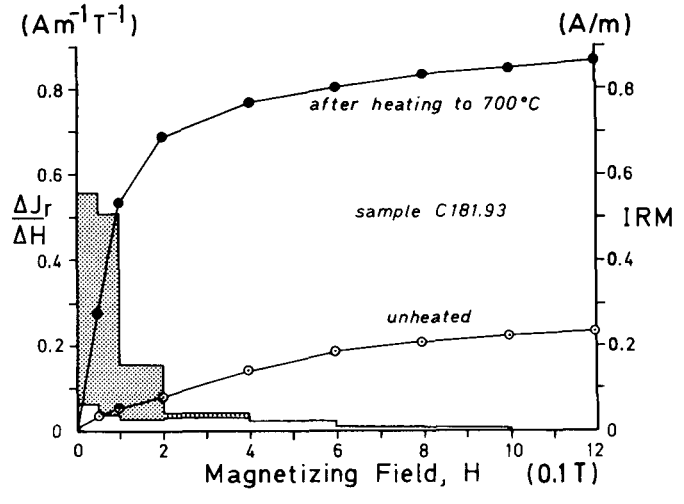
Age	Number of samples	Decl.	Incl.	α_{95}	k
Oligocene					
Normal	47	337.8	46.8	4.1	26.11
Reversed	75	156.8	-43.9	4.1	16.57
Total	122	337.2	45.0	3.0	19.31
VGP position: (1) 65.0N 247.3E, dp = 2.4, dm = 3.8 (2) 65.5N 249.3E, α_{95} = 3.1					
Eocene					
Normal	185	330.9	41.0	2.6	16.77
Reversed	204	137.6	-40.6	2.9	12.32
Total	389	324.0	41.0	2.0	13.45
VGP position: (1) 54.3N 260.0E, dp = 1.5, dm = 2.4 (2) 55.4N 261.8E, α_{95} = 2.0					
Paleocene					
Normal	17	330.3	44.4	12.2	9.47
Reversed	30	134.3	-35.2	12.0	5.76
Total	47	319.8	39.0	8.8	6.52
VGP position: (1) 50.5N 262.4E, dp = 6.3, dm = 10.5 (2) 51.3N 264.3E, α_{95} = 8.8					

Note: paleomagnetic directions and VGP calculated from (1) analysis of directions and (2) analysis of poles.









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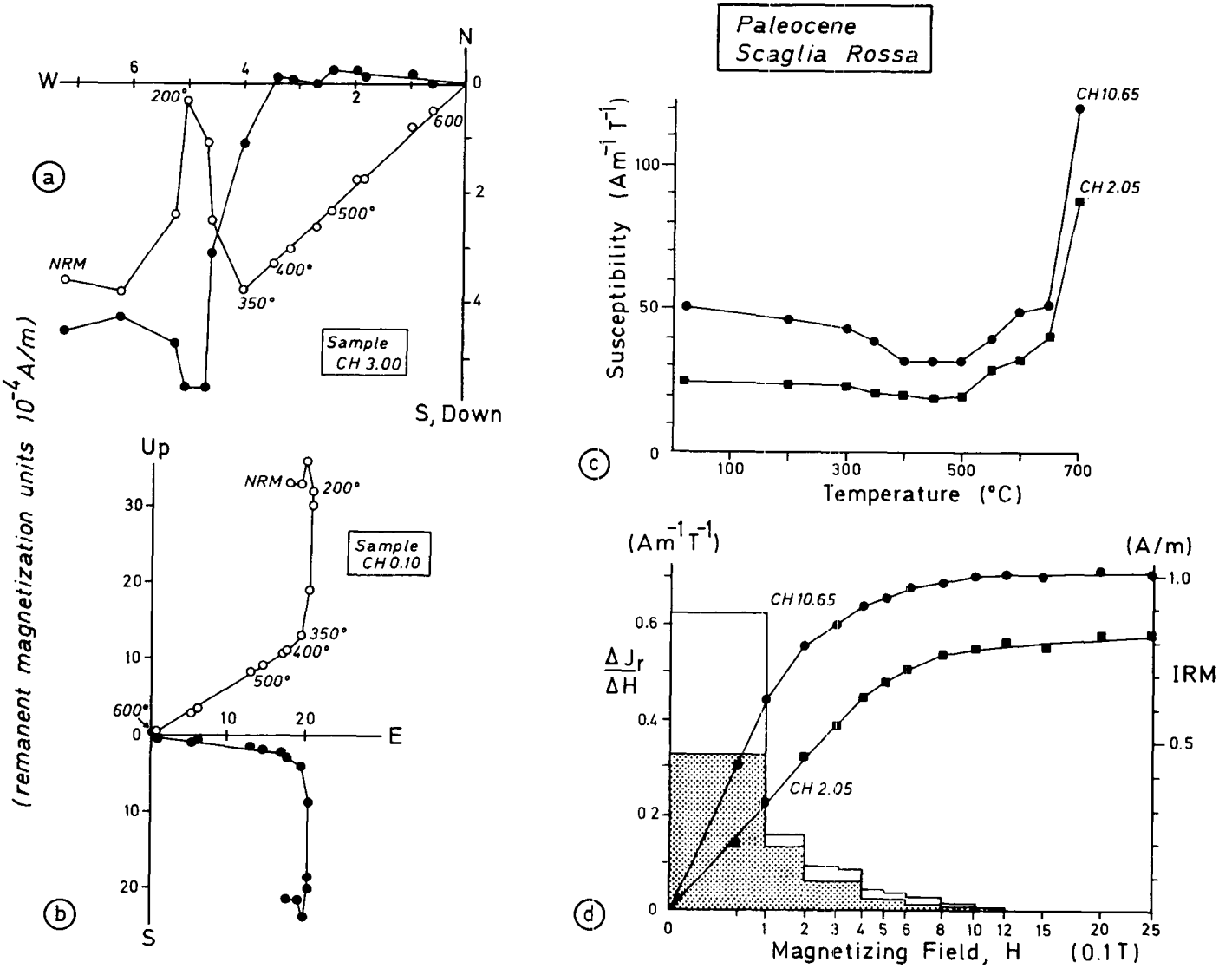


Figure 12. Thermal demagnetization of the dark red Paleocene Scaglia Rossa removes components parallel and antiparallel to the present (Holocene) field below 350 °C. Above this temperature the ChRM is again clearly defined. The susceptibilities show noticeable decrease after heating to 300–400 °C and a sharp increase after heating to 550 °C. The dominant magnetic minerals in unheated samples were again deduced from the IRM curves and coercivity spectra to be magnetite and hematite.

at 300 °C, 400 °C, and 500 °C; and Scaglia Rossa samples at 350 °C, 400 °C, 450 °C, and 500 °C. Additional thermal demagnetization in 25-degree steps to 600 °C was carried out where necessary. The ChRM was taken as the first measurement after the direction stopped changing. Susceptibility was monitored after each heating, and remanence measurements were discontinued if an appreciable susceptibility increase was accompanied by increasing directional scatter. ChRM intensities averaged 34×10^{-5} A/m, 6×10^{-5} A/m, and 10×10^{-5} A/m in the Scaglia Cinerea, Scaglia Variegata, and Scaglia Rossa, respectively.

The ChRM direction was checked by comparison of twin specimens in samples where these were available. The ChRM vectors represent both normal and reversed polarities, which form well-clustered, almost antipodal groups (Fig. 15). More than one-half of the samples (55%) in each formation are reversely magnetized. The variation of bedding attitude was too small to permit a meaningful tectonic fold test. Correcting for the 50° to 60° northeasterly dip of the bedding by rotations around an assumed horizontal fold axis produced no significant changes in the statistical precision parameter k (Fisher, 1953). However, the west-east direction groups moved to northwest-southeast azimuths, which are typical for this part of Umbria.

The paleomagnetic directions from the three sections were grouped according to age and averaged statistically (Fisher, 1953); virtual geomagnetic pole positions were calculated for each stage (Table 1). The average inclination for each stage is generally 5° to 10° flatter than expected in Umbria from either European or African paleomagnetic poles (Van der Voo and French, 1974; Irving, 1977). The Paleocene and Eocene declinations are rotated counterclockwise by about 40° relative to "European" directions

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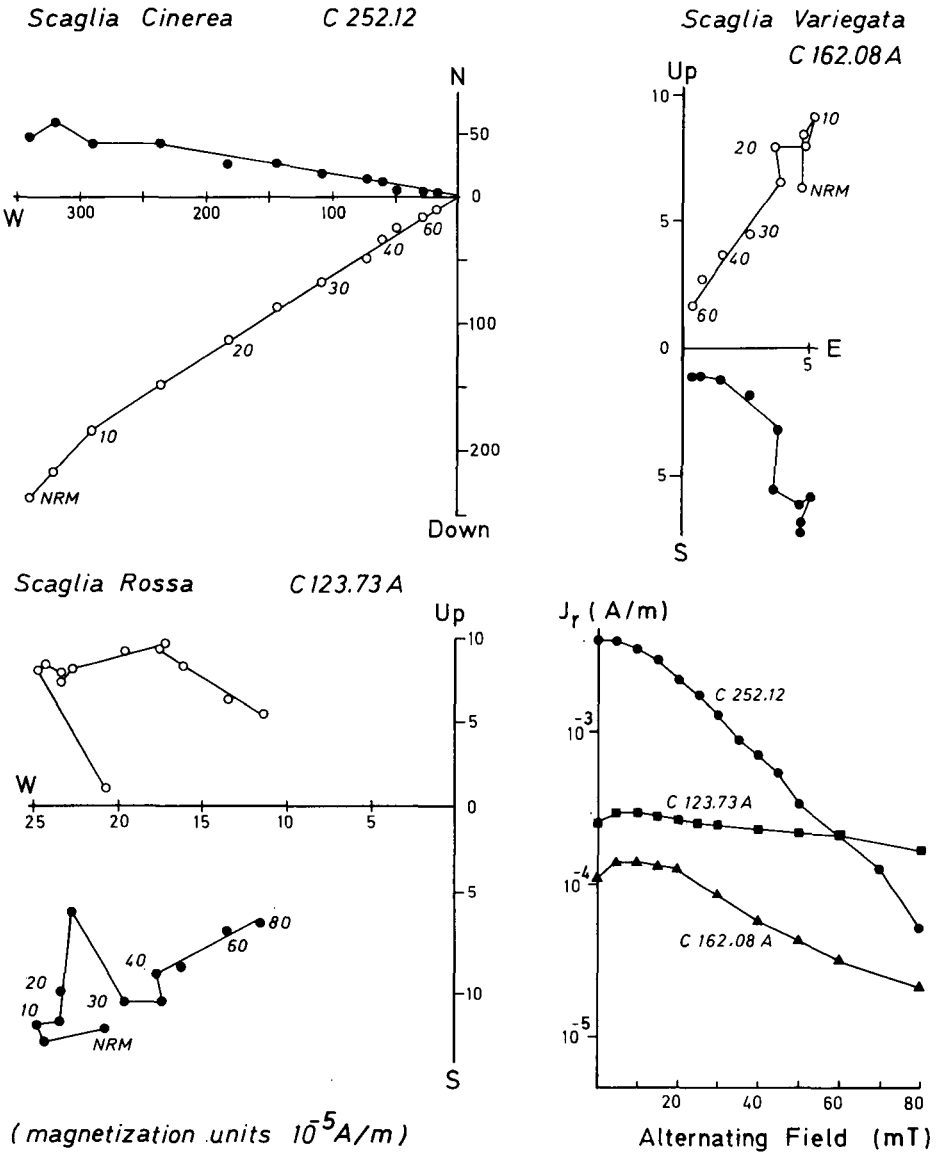


Figure 13. Alternating Field (AF) demagnetization achieves complete cleaning of the Scaglia Cinerea. In the Variegata and especially the Scaglia Rossa, use of high fields defines a vector which decreases toward, but does not reach, the origin. Numbers on the vector diagrams indicate peak field in mT; open circles are in the vertical east-west plane, and solid circles are in the horizontal plane, prior to tectonic correction.

and by about 30° relative to "African" directions. The Oligocene directions are also rotated counterclockwise, but by smaller amounts; the rotations are about 25° relative to "European" and about 20° relative to "African" reference directions, respectively.

Previous studies in Umbria have demonstrated that large-scale tectonic conclusions cannot be drawn safely from a single paleomagnetic section. Local and regional variations of fold axis orientation have strongly perturbed Umbrian paleomagnetic declinations (Channell and others, 1978). By comparison with directions from the Bottaccione gorge (Roggenthen and Napoleone, 1977; Napoleone and others, in press) only 2 km to the southeast (Fig. 1a), the Contessa Paleocene and Eocene declinations are rotated some 10° further counterclockwise. However, the Contessa directions are consistent with what is known about the geodynamic history of Umbria.

The Umbrian sequence was deposited on the continental margin of the Adriatic promontory of the African plate (Channell and others, 1979b), Umbria moved northward and rotated counterclockwise as part of the African plate until the late Tertiary, when it underwent ~25°-counterclockwise rotation relative to Africa, either accompanying rotation of an Italian microplate (VandenBerg, 1980) or as the result of décollement on Triassic evaporites (Channell and Tarling, 1975). The trends of paleomagnetic directions from the Contessa sections are compatible with this model and suggest that the final rotation took place in the Neogene.

MAGNETIC STRATIGRAPHY

The ChRM directions define discrete zones of uniform inclinations and declinations when plotted against stratigraphic position (Figs. 3, 5, 7). The magnetic polarity reversals are marked by synchronous changes in declination and inclination. The remanent intensity shows clear indication of being lithologically controlled. It decreases by about an order of magnitude in the vicinity of the gradual transition from Scaglia Cinerea to Scaglia Variegata near 220 m in the Quarry section (Fig. 7), and 160 m in the Highway section (Fig. 3). Just below the covered gap in the Quarry section the intensity reaches a peak which correlates with high intensities above the marker bed sequence L–Q in the other two sections. Other intensity peaks occur near the white marker and K beds in the Road and Highway sections (Figs. 3, 5). This lithological control over the remanent intensity reflects the varying concentration of magnetic mineral in the limestones.

The declination data contain the effects of the Umbrian tectonic rotation. These were removed by subtracting the declination of the mean ChRM direction in each section from the individual declinations. The site latitude during deposition was deduced from the inclination of the mean ChRM direction and, together with the individual inclinations and corrected declinations, was used to compute a relative latitude for the virtual geomagnetic pole (VGP). Defined in this way, the VGP latitude should be 90N for normal polarity and 90S for reversed. It served as a convenient single parameter from which the magnetic polarity stratigraphies of the three sections were determined (Figs. 4, 6, 8). Most polarity

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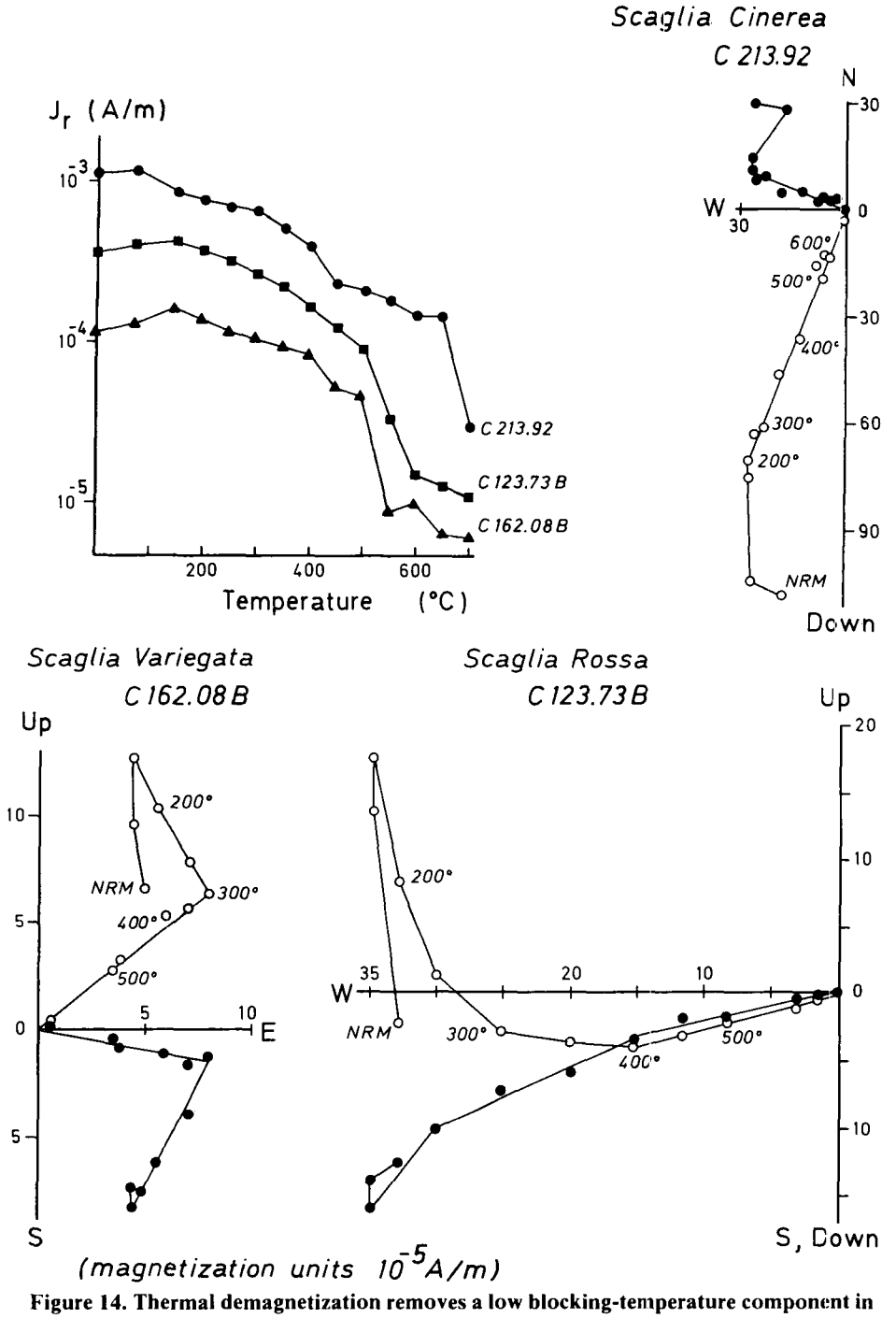


Figure 14. Thermal demagnetization removes a low blocking-temperature component in each lithology. The characteristic remanent magnetization (ChRM) direction is defined above 200 °C in the Scaglia Cinerea, 300 °C in the Scaglia Variegata, and 400 °C in the Scaglia Rossa.

zones are clearly defined and are represented by several samples each. However, the chosen sample spacing was too wide to describe the shorter magnetozones completely; this was particularly unfortunate in the Quarry section, which could not be resampled later.

BIOSTRATIGRAPHY

The biostratigraphy of the Scaglia Cinerea has been studied by several workers. Bonarelli (1901) considered the unit to be Upper Cretaceous on the basis of rare macrofossils; but these forms, notably the giant Lima *Acesta,* are poor age indicators (Carole S. Hickman, 1979, personal commun.), as is emphasized by the fact that Renz (1936; 1951, p. 38) considered them surely Miocene! More reliable planktic foraminiferal dating locates the gradational Scaglia Cinerea–Scaglia Variegata contact in the vicinity of the Eocene-Oligocene boundary. The foraminiferal zonation has been progressively refined by Renz (1936, 1951), Luterbacher and Premoli Silva (1962), Cati (1964), Baumann and Roth (1969), and Baumann (1970). Nannoplankton studies further confirm this dating (Baumann and Roth, 1969; Roth, 1970).

Foraminiferal zonations of the Scaglia Rossa have been carried out by a number of specialists. The paleontological dating of the Bottaccione gorge (Renz, 1936, 1951; Luterbacher and Premoli Silva, 1962, 1964; Premoli Silva, 1977) provided the biostratigraphic framework for the magnetostratigraphic correlations of the Paleocene and Late Cretaceous oceanic magnetic anomalies. Different sites have been dated paleontologically in connection with paleomagnetic studies in Umbria (VandenBerg and others, 1978; Channell and others, 1978) and magnetostratigraphy in the Southern Alps (Channell and others, 1979a).

In the present magnetostratigraphic study, the Contessa Quarry and Road sections were dated using planktic foraminifera from washed samples (M. Toumarquine) and also by calcareous nannofossils (K. Perch-Nielsen). The Contessa Highway section was dated most reliably from planktic foraminifera studied in thin sections (I. Premoli Silva).

The Contessa sections contain an abundant microfauna and microflora. The major problem encountered in studying planktic foraminifera and calcareous nannofossils was the generally poor preservation. The planktic foraminifera were particularly poorly preserved in all samples from the top of the middle Eocene to the base of the middle Oligocene; the lower Oligocene could not be zoned. Above and below these intervals, the preservation was somewhat better. All the foraminiferal zones from the beginning of the Paleocene to the base of the Miocene have been recognized in these sections, although in some parts neighboring paleontological zones were not differentiated and several zonal boundaries could be determined only approximately.

The sediments are lithified nannofossil oozes. The planktic foraminifera are dis-

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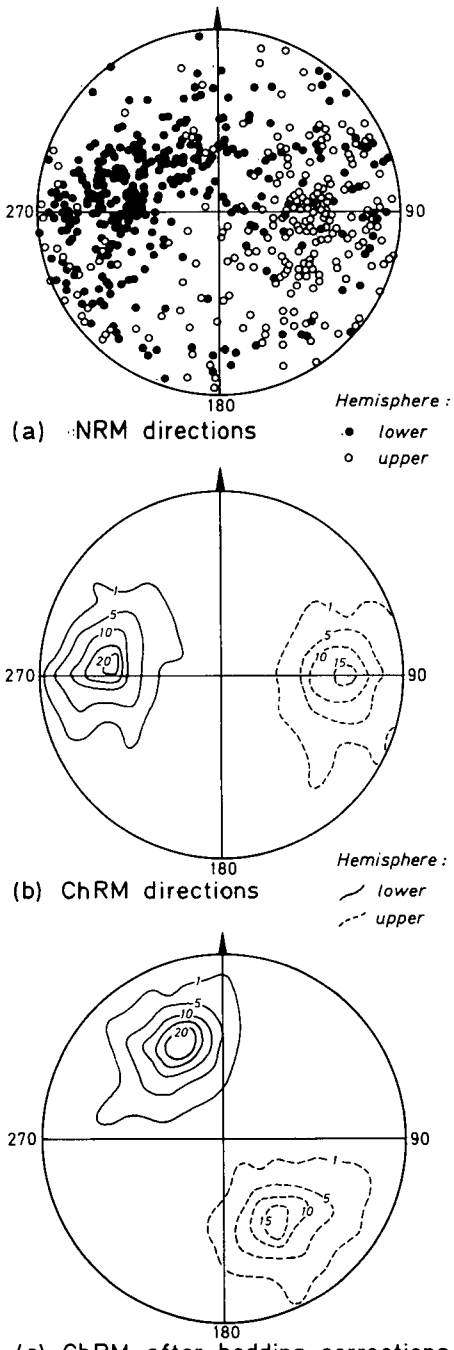


Figure 15. Stereographic plots of (a) individual sample NRM direction, and equal-area density diagrams of the cleaned ChRM directions (b) before and (c) after bedding corrections. Numbers on equal-area contours indicate point density as percentage per 1% area.

persed in a fine matrix consisting of recrystallized and overgrown calcareous nannofossils which form a cryptic groundmass. Red layers, particularly in the Scaglia Variegata, display a large number of solution seams which have dissolved the outer portions of foraminiferal tests; this may explain the poor preservation of foraminifera in the washed residues. Reworking, occasionally of Late Cretaceous but more commonly of Paleocene faunas, is found in red layers of Scaglia Variegata.

The zonation of Martini (1971), which is based on land sections, was used for the calcareous nannofossils (NP, NN zones). For sediments younger than middle Eocene, the zones of Bukry (1975) were also used, with the new designations (CP, CN) of Okada and Bukry (1980); these zones are based mainly on low-latitude, open-ocean sequences. Although sometimes species could not be determined due to the heavy overgrowth, the calcareous nannofossil data complemented the planktic foraminiferal observations and reinforced the interpreted locations of the stage and epoch boundaries in the Quarry and Road sections (Figs. 6, 8).

Particular difficulties with the nannofossil zonation were encountered in some places. For example, in the Road section (Fig. 6), two samples were assigned to zone NP10 despite the fact that the species defining this zone was not found. The assignment is based on the absence of any *Fasciculithus* (a genus which usually disappears near the Paleocene-Eocene boundary) and the absence of *Tribrachiatus orthostylus* (the first occurrence of which approximates the base of the following zone NP11). In the Quarry section (Fig. 8), the Oligocene nannofossil zonation is unreliable because of poor preservation. The last occurrence of *Reticulofenestra umbilica* seems unreliable; the species is absent intermittently in many middle Eocene through Oligocene samples. The last occurrence of *Isthmolithus recurvus* is used instead of the last occurrence of *R. umbilica* to define the base of CP17 and NP23. The last occurrence of *Sphenolithus distentus* is found in this section below the first occurrence of *S. ciperoensis*, which is opposite to the normal sequence. Since these events normally bound zone NP24, this nannofossil zone cannot be defined for the Quarry section. The stratigraphically important sphenoliths are very rare throughout the Oligocene. *Discoaster druggi*, the first appearance of which marks the base of Miocene zone NN2, was not found.

In the Quarry section (Fig. 8), some zones (for example, the uppermost Eocene coccolith zones NP19/20 and the corresponding *G. cerroazulensis* foraminiferal zone) are unusually thin. This may indicate reduced sedimentation rate; alternatively, a fault may have cut out part of the section. The many small faults from 222.35–256.3 m lie in an interval which could not be zoned with foraminifera because of their poor preservation, but in which a nannofossil zonation was possible. These faults appear not to have greatly disturbed the magnetic polarity sequence.

Absence of paleontologically useful marly horizons in the Road section resulted in a gap in the zonation and in insecure determinations of zonal boundaries in certain intervals (Fig. 6). The base of the *H. aragonesis* zone and the top of the *G. aragonensis* zone could not be determined. The intervening *G. pentacamerata* zone was not determined and undoubtedly lies in the unsampled gap. Therefore, the boundary between early and middle Eocene in this section could not be precisely located, but its inferred position relative to lithostratigraphic and magnetostratigraphic markers is compatible with the locations of this boundary in the Contessa Highway (Figs. 2, 4) and Bottaccione sections (Fig. 2).

In the Highway section (Fig. 4), free specimens obtained from washable material were poorly preserved, particularly in the middle and lower parts of the section, and were considered less reliable than thin sections cut from the samples used for magnetic measurements. The sampling in some instances did not represent full coverage and the boundaries between some of the zones are uncertain. However, the section appears to be continuous biostratigraphically. All the foraminiferal zones from the earliest Paleocene *G. eugubina* zone up to the late Eocene have been recognized. At about 31 m, the lower part of the *G. velascoensis* zone is missing, possibly because of slumping.

CORRELATIONS BETWEEN SECTIONS

Lithologic correlations between the Contessa Road and Highway sections and the Bottaccione section of Napoleone and others (1982) can be made at several levels (Fig. 2). The base and top of the upper cherty member of the Scaglia Rossa, the white marker bed about 12–20 m higher than this member, and the marly triplet K are found in all three sections. The six marker beds L–Q were found in the Contessa Road and Highway sections; these were absent in the Bottaccione section, and the absence of a related magnetozone cor-

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responding to anomaly 19 leads us to conclude that about 6 m of section is missing near the top of the Bottaccione section. The sequence L–N could also be identified in the Gully segment of the Quarry section, and here a distinctive magnetozone "fingerprint" confirms correlation to the other sections.

Biostratigraphic and magnetostratigraphic data agree well with the lithostratigraphic correlations. For convenience, the positive magnetozones are identified with the corresponding oceanic magnetic anomaly numbers (Fig. 9). The base of the *G. cerroazulensis* zone, the positions of the *G. semiinvoluta–T. rohri* and *G. lehneri–G. subconglobata* zonal boundaries, and the distinctive "fingerprint" appearance of the magnetozone sequence 18/19/20 serve as critical features for correlating the lower part of the Quarry section and the upper parts of the other sections (Fig. 2). Additionally, the narrow zone of the distinctive fauna *O. beckmanni* brackets the lithological marker Q and the base of magnetozone 18 in the Road and Highway sections. The base of the *G. cerroazulensis* zone is in a positive magnetozone in the Contessa Highway and Bottaccione sections, but in the negative zone just below magnetozone 15 in the Quarry section. Poor faunal preservation makes precise boundary location difficult, but an undetected fault may have cut out part of the Quarry section at this level. The relationships of the magnetozone sequences 21–25 to the K-triplet, the white marker, and the upper cherty member, as well as to the *H. aragonesis–G. pentacamerata, G. formosa formosa–G. subbotinae*, and *G. edgari–G. velascoensis* faunal boundaries, enable correlation of the Road, Highway, and Bottaccione sections at additional key levels (Fig. 2).

COMPARISON OF CONTESSA MAGNETIC STRATIGRAPHY WITH OCEANIC MAGNETIC ANOMALY REVERSAL SEQUENCE

In revising the magnetic time scale, LaBrecque and others (1977) incorporated modifications from several preceding investigations and altered the relative position of anomaly 4A and the spacings of anomalies 29 to 34. They dropped the apparent reversal sequence of anomaly 14, as it did not occur in most marine magnetic profiles. Otherwise, the Paleogene part of their time scale was based on the assumption of a constant rate of sea-floor spreading in the South Atlantic. It serves here as a basis of comparison with the Contessa magnetostratigraphic sections.

The Contessa Road and Highway sections were combined to form a composite magnetic stratigraphy for the lower Paleogene; the Contessa Quarry section above the magnetozone 15 represented the upper Paleogene. These segments were then matched in optimum positions against the LaBrecque and others (1977) oceanic magnetic reversal sequence (Fig. 9). The Contessa sections contain fewer reversal boundaries than the oceanic record. This may be due to the sample spacing being too wide to detect very short polarity events, or to these short events being poorly recorded by the postdepositional magnetization process in these pelagic limestones. The tie-lines indicate minor fluctuations in ocean-floor spreading rate and/or limestone sedimentation rate. However, the magnetostratigraphic correlation is generally very good and allows clear association of magnetozone with corresponding oceanic magnetic anomaly. It determines the positions of the major epoch boundaries and some of the stage boundaries within the oceanic reversal sequence. These identifications alter, sometimes substantially, the boundary locations inferred by linear interpolation in earlier magnetic time scales. With respect to the time scale of LaBrecque and others (1977), the most notable changes are the following:

1. **The Oligocene-Miocene boundary**, determined by planktic foraminifera and also by calcareous nannofossils, lies in the interval of negative polarity just older than anomaly 6C. This position differs from the time scale of LaBrecque and others (1977), who interpolated the boundary below anomaly 6B, but agrees closely with the relocation by Ness and others (1980).

2. **The Eocene-Oligocene boundary**, defined again by planktic foraminifera and calcareous nannofossils, lies in the negative interval between anomalies 13 and 15. LaBrecque and others (1977) interpolated this boundary between anomalies 15 and 16; Ness and others (1980) interpolated it within anomaly 16.

3. **The Paleocene-Eocene boundary**, defined in the Contessa Road section by both planktic foraminifera and calcareous nannofossils and in the Contessa Highway section and the Bottaccione section (Napoleone and others, in press) by planktic foraminifera, lies in the negative polarity interval between anomalies 24 and 25 (Figs. 2, 9). LaBrecque and others (1977) interpolated this boundary within anomaly 23, and Ness and others (1980) interpolated it within anomaly 24. The new location agrees well with recent magnetostratigraphic results of Butler and others (in press) on continental sedimentary sequences from the western United States dated by land-mammal fossils.

In addition to the above locations of the major Paleogene epoch boundaries, our results also permit direct determination of several stage boundary locations. The transition from middle to late Oligocene is at the young end of anomaly 9, the middle-late Eocene boundary is near the young end of anomaly 18, and the early-middle Eocene boundary lies near the young end of anomaly 22.

DISCUSSION AND CONCLUSIONS

The magnetostratigraphic results from the Contessa sections, and complementary results from the Bottaccione section (Napoleone and others, in press), have provided independent confirmation of the Paleogene geomagnetic polarity history first deduced from oceanic magnetic anomalies. Although some of the paleontological zonation was obscured by extreme poverty of faunal preservation in some sections, the combination of data from all sections gives a satisfactorily well-dated magnetic stratigraphy. The Paleogene magnetic reversal sequence can now be correlated with the biogeochronological time scale at several quite well-defined tie-points. The most pronounced change resulting from the present study and that of Napoleone and others (1982) concerns the new location for the Paleocene-Eocene boundary in the negative interval just younger than anomaly 25.

The radiometric ages of the Paleogene epoch (and some sub-epoch) boundaries, compiled by Hardenbol and Berggren (1978) and revised by Ness and others (1980) using new decay constants and abundances, are shown where applicable in Figure 9. If the epoch boundary ages are accurate, the assumption of linear sea-floor spreading in the South Atlantic inherent in

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earlier magnetic time scales (Heirtzler and others, 1968; LaBrecque and others, 1977; Ness and others, 1980) may not be valid. Alternatively, if the constant-spreading hypothesis is correct, the radiometric ages associated with the stage boundaries may require adjustment. Comparison with the scale of LaBrecque and others (1977) indicates comparatively little change in Oligocene spreading rates. Eocene spreading rates were 25% to 30% faster, and Paleocene spreading rates 20% to 30% slower, than rates given by the time scales of LaBrecque and others (1977) or Ness and others (1980). The magnetostratigraphic correlations also affect estimates of the duration of episodes of constant sea-floor spreading and the timing of events such as changes in spreading rate or direction.

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